Monitoring Internal Load Parameters During Simulated and Official Basketball Matches

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Abstract

Moreira, A, McGuigan, MR, Arruda, AFS, Freitas, CG, and Aoki, MS. Monitoring internal load parameters during simulated and official basketball matches. *J Strength Cond Res* 26(3): 861–866, 2012—The purpose of this study was to compare the internal load responses (session rating of perceived exertion [RPE] and salivary cortisol) between simulated and official matches (SM and OM). Ten professional basketball players participated in 2 OMs and 2 SMs during the competition season. Subjects provided saliva samples 30 minutes before the prematch warm-up (PRE) and 10 minutes after the end of the match. Session RPE (CR-10 scale) was assessed 30 minutes after each match. The results from the 2-way analysis of variance showed significant differences for post-OM salivary cortisol as compared with pre-OM values ($p < 0.05$). No changes were observed for cortisol during the SM. Before the OM, a significant difference in salivary cortisol was observed as compared with pre-SM values ($p < 0.05$). Moreover, the OM session RPE was significantly greater than that of SM. There was a significant correlation between session RPE and cortisol changes ($r = 0.75$). In summary, the results of this study showed a greater magnitude of cortisol and session RPE responses after OM as compared with that after SM confirming the hypothesis that a real competition generates a greater stress response than a simulated condition does. The anticipatory effect was also observed in the OM. In addition, the results indicate that session RPE seems to be a viable tool in monitoring internal loads, and the results are useful in providing a better understanding of internal loads imposed by basketball training and competitions. The precise monitoring of these responses might help the coaches to plan appropriate loads maximizing recovery and performance.

Key Words team sports, saliva, ratings of perceived exertion, cortisol

Introduction

The main goal for coaches is to maximize athletic performance. To achieve this goal, it is critical to prescribe an optimal training load, with appropriate recovery periods to induce positive outcomes, and therefore enhance performance. Therefore, it is important that methods of monitoring precise training loads are developed. Basketball matches impose significant physiological stress on the athletes (13,15,17). Recently, Manzi et al. (13) reported that professional basketball imposes great physiological and psychological stress on players through training sessions and official competitions.

Dhabhar and McEwen (3) provided an integrated definition for the concept of stress. The authors pointed out that "stress is a constellation of events, consisting of a stimulus (stressor) that precipitates a reaction in the brain (stress perception), that activates physiological fight-or-flight systems in the body (stress response)." The hypothalamic-pituitary-adrenal (HPA) axis is a major part of the neuroendocrine system and, in conjunction with the sympathetic system, connects the brain with the periphery of the body (18). The HPA axis function can be assessed by means of salivary cortisol responses (24).

The rating of perceived exertion (RPE) can be used to assess the training load imposed by exercise (1,6,10,14). Foster (6) proposed a simple method to calculate the magnitude of training load using the (CR-10) scale (1,10). However, no previous study has compared the stress response between simulated matches (SMs) and official matches (OMs) in professional basketball players using the session RPE method associated with the salivary cortisol responses. Manzi et al. (13) monitored the magnitude of training load by means of session RPE and 2 heart rate–based (HR) methods. These authors pointed out the applicability of the session RPE method in quantifying the training load during

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26(3)/861–866

Journal of Strength and Conditioning Research

distinct training types of exercise; however, during this study, the session RPE method was not tested during official matches.

In a practical sport setting, the use of markers that precisely reflect the magnitude of the stress related to different conditions and environments could help coaches to better understand how the athletes are coping with stressors, thus providing valuable information to optimize the training periodization. For example, comparing the response of athletes with training sessions with real competition outcomes by means of RPE (perceptual marker) and cortisol (physiological marker) responses emerges as an interesting approach.

The behavior and the association between these markers (RPE and salivary cortisol concentration) could be distinct between training and competition settings. The competition setting may be characterized as a situation that induces ego involvement, novelty, and unpredictable effects and outcomes, leading to negative affective states (2). In this situation, it is reasonable to assume that the greater level of stress is imposed (7) on the athletes leading to a greater effect on cortisol levels and greater RPE responses.

Therefore, the purpose of this study was to investigate and compare the stress responses (assessed by Foster's session RPE method and salivary cortisol changes) between SMs and OMs in elite professional basketball players. It was hypothesized that greater stress responses would be observed after official basketball matches as compared with those observed after simulated training matches. It is also hypothesized that there is an association of these indicators during OM revealing a consistent pattern of the interaction between stress perception and stress response in the sports competition setting.

Methods

Experimental Approach to the Problem

The study group was tested across 4 separate occasions (i.e., 2 OMs and 2 SMs) across 4 weeks, during the competition basketball season. Both official games were played between 20:00 and 22:00 hours, approximately. The simulated matches were designed to replicate an official game and were played at the team's training facility under the actual playing rules and at the same time as that of the OMs. The simulated games involved 5 starting players in each team and consisted of four 10-minute quarters with an interval of 2 minutes between each quarter. Each game was preceded by a 30-minute warm-up comprising light aerobic exercise, basketball and team drills, and dynamic stretching of the major muscle groups. All of the 10 investigated players participated in the 4 matches. Because of prior scheduling of training and competition, the SMs were conducted before the OMs (Figure 1). In both competition settings, the players were encouraged to drink water between break periods to maintain hydration levels. Food intake on each day was not strictly monitored during this study, but players were instructed to maintain their normal dietary intake on each day of testing. The session RPEs after each match were obtained from all the players involved in the study. Saliva samples were also collected before and after each match.

Subjects

Ten elite male basketball players volunteered for this study (mean ± *SD*: age, 26.4 ± 3.8 years; height, 196 ± 10 cm; and body mass, 100 ± 14 kg). They played for a professional team competing for the State Basketball Championship (São Paulo, Brazil). On a weekly basis, each player was typically training twice a day (90–120 minutes per session), 6 $d \cdot wk^{-1}$ and played in 1 or 2 OMs. The training sessions consisted of basketball drills, tactics, sprints, intermittent running exercises, and specific conditioning work, and weight training and plyometrics. The participants provided informed consent before the study commenced. All procedures received local ethics committee approval.

Saliva Collection and Analysis

The subjects provided saliva samples 30 minutes before the prematch warm-up (PRE) with postmatch (POST) samples being collected within 10 minutes of the completion of each game. The samples were collected in sterile 15-ml tubes and stored at −80° C until they were assayed. The subjects abstained from consuming food and caffeine products for at least 2 hours before the collection of saliva (8). Salivary cortisol concentrations were measured in duplicate by means of an enzyme-linked immunosorbent assay (DSL ACTIVE Cortisol EIA Kit–10-67100–Diagnostic Systems Laboratories, Inc., Webster, TX, USA) using previously described methods (16). Briefly, saliva samples were thawed and centrifuged at 1,630*g* for 20 minutes. Then, 25 μL of calibrators, control, and supernatant saliva samples were added to microtiter wells and, in sequence, 100 μL of enzyme conjugate solution and 100 μL of the

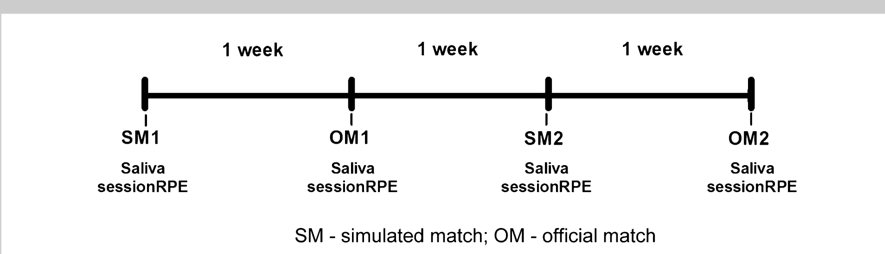

Figure 1. Investigation time line.

cortisol antiserum were added to each of the wells and incubated for 45 minutes, with constant shaking (500 rpm), at room temperature (25° C). After incubation, the plate was aspirated and washed 5 times with the Wash Solution using an automatic microplate washer. Then, 100 μl of tetramethylbenzidine substrate solution was added, and the plate was incubated on a plate shaker for 15 minutes at room temperature. Finally, 100 μl of the stop solution was added to the wells, and the absorbance was read on the plate reader at 450 nm (ELX 800VV–Universal Microplate Reader, Bio-TeK instruments, Winooski, VT, USA). From a calibration curve (absorbance vs. cortisol concentration of the calibrators), the concentration of cortisol in each sample was interpolated. The coefficient of variation of the intra-assays was 4.8%.

Match Internal Load

The internal load for each athlete was calculated according to the methods of Foster (6). This method of monitoring training and competition loads required each athlete to provide an RPE using the CR-10 scale (Table 1) for each exercise session along with a measure of session time. To assess the match intensity, after 30 minutes of the end of the match, the athletes were asked a simple question: "How was your workout?" and a chart was shown that outlined the full RPE scale with the appropriate explanations. The data were recorded by the same investigator on all the occasions.

A single number representing the magnitude of match load was then calculated by multiplying the intensity by its duration (Match internal load = Session RPE × total match duration in minutes). We considered the real duration of the match, in minutes, for all players; the duration was recorded from the start to the end of the game including all stops, as used by Manzi et al. (13). The data were recorded by the same investigator on all occasions, and the study group was familiarized with the use of session RPE before the beginning of the experiment.

Table 1. Modified CR-10 scale.

Rating	Descriptor
0	Rest
1	Very, very easy
2	Easy
3	Moderate
4	Somewhat hard
5	Hard
6	
7	Very hard
8	
9	
10	Maximal

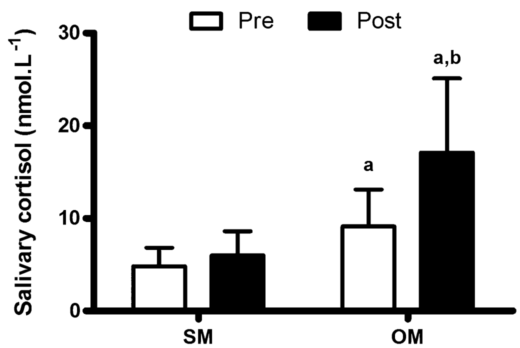

Figure 2. Cortisol responses to simulated matches (SMs) and official matches (OMs); a = significant difference between SM and OM; b = significant increase for post-OM cortisol as compared with the pre-OM value.

Statistical Analyses

Data distribution was assessed using the Shapiro-Wilk test. Initially, a 2-way analysis of variance (ANOVA) was used to compare the salivary cortisol levels before and after each SM and OM. No significant differences between SM1 and SM2, and between OM1 and OM2 were observed. Therefore, a 2-way ANOVA) with repeated measures (moment) was used to compare the salivary cortisol levels averaged across the 2 SMs and the 2 OMs (condition [SMs vs. OMs] and moments [premoment and postmoment]). The Tukey Honestly Significant Difference (HSD) test was used as the post hoc procedure. In the case of violation of the assumption of sphericity, the significance was established by using the Greenhouse-Geisser procedure. A paired t-test was used to compare the perceptual match load with that of SMs and OMs. The relationship between the absolute changes in

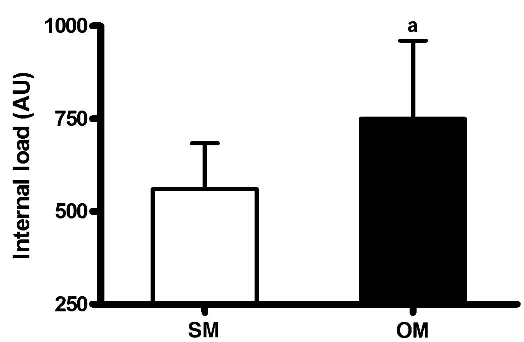

Figure 3. Internal load imposed by simulated matches (SMs) and official matches (OMs) assessed by the session rating of perceived exertion (RPE) method; a = significant difference between SM and OM ($p < 0.01$).

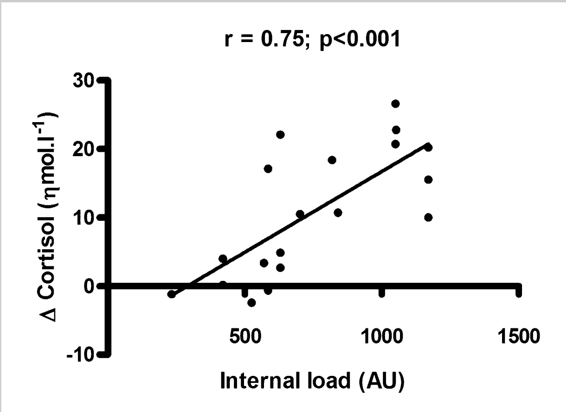

Figure 4. Correlation between pre-post cortisol changes and session rating of perceived exertion (RPE) internal loads.

salivary cortisol levels (differences from PRE to POST values) and match internal load during OM was verified by means of Spearman correlation coefficients. The level of significance was set at $p \leq 0.05$.

Results

Figure 2 shows the cortisol responses to SM and OM. No changes were observed for cortisol during SM. A significant difference was observed between pre-OM and pre-SM ($p <$ 0.03) and between post-OM and post-SM ($p < 0.01$). A significant increase was also detected for post-OM cortisol as compared with the pre-OM value ($p < 0.01$).

Figure 3 shows session-RPE internal loads induced by SM and OM. The OM internal load was significantly greater than the SM internal load ($p < 0.01$).

Figure 4 shows that there was a significant correlation between match internal load and cortisol changes during OM ($r = 0.75$, $p < 0.01$).

Discussion

The findings of the present investigation show that competition constitutes a real stress model and, in turn, leads to increases in the internal load markers (e.g., salivary cortisol and session RPE). The results showed that there was a greater magnitude of cortisol and session-RPE responses after OM than after SM. This confirms the hypothesis that a real competition generates a greater stress response than simulated conditions do, even when the simulated matches were designed to replicate an official game under the actual playing rules.

These findings suggest that if the aim of SMs is to replicate several aspects encountered in the competition environment, it is plausible to admit that, at least to internal load parameters assessed, this aim is difficult to achieve. It could be that elite athletes might be less responsive to regular training stimuli (e.g., SMs) because of their high level of experience. Moreover, this result may be explained, at least in part, by greater physical and psychological demands induced by OM as compared with those induced by SM. It is reasonable to speculate that greater demands were imposed during the OM. The elevated cortisol responses during the competition could be viewed as a positive effect, because cortisol plays a key role in regulating energy availability from fat, carbohydrate, and protein resources (25).

A rise in cortisol levels has been observed in many sports competitions (4,5,9,11,12,19–22) and, in most cases, these changes appeared to enhance the performance and behavioral outcomes. For example, improvements in the performance of Olympic weightlifters were noted during actual (vs. simulated) competition when salivary cortisol concentrations also increased more than twofold (20). Judoists displaying higher salivary cortisol levels also had higher motivation to perform and obtained better outcomes (21), whereas higher cortisol levels were found in judo winners when compared with those in losers (23). Thus, acute elevations in cortisol may benefit competitive performance for some athletes.

In addition, an inherent competitive-unpredictable situation associated with high-intensity physiological stressful stimulus seems to facilitate the HPA axis response, even in professional players who are accustomed to stressful competition and consequently exposed repeatedly to this kind of stimulus.

The well-known anticipatory effect was also observed in this study, comparing prevalues with those of training and the OM. Other researchers have consistently demonstrated this effect using different sporting modalities. For example, Filaire et al. (5) found higher resting cortisol concentration during competition days compared with that during control days, and higher values before the first interregional match compared with that measured before the first regional competition match. The results of this study are in agreement with the assumption that an official and important match could induce greater initial cortisol concentrations because of the anticipatory effect.

Session RPE is a simple and reliable marker of internal load in team sports (1,10,13). Manzi et al. (13) observed significant relationships between individual session RPE and all individual HR-based training loads ($r = 0.69$–0.85; $p <$ 0.001) for training and competition in elite basketball players. Strong correlations between session-RPE training load and HR-based training load were also observed for soccer as previously reported by Impellizzeri et al. (10) and Alexiou and Coutts (1).

As previously hypothesized, it was detected as a significant correlation between session RPE and cortisol responses during competition. Session RPE responses were associated with cortisol responses ($r = 0.75$), indicating the validity of the RPE method for monitoring internal load in sports competition, particularly within professional basketball players. It is important to note that, as pointed by Manzi et al. (13), the applicability of the session RPE method to quantifying exercise training load during distinct conditions (e.g., training and competition) had not been previously tested in professional basketball players.

However, despite the strong correlations between session RPE and other physiological parameters verified by some authors (1,10,13), this study seems to be the first one to investigate the responses from session RPE in professional basketball players during OMs. In addition, we noted that session RPE method is not only a useful method for monitoring the internal load in a practical setting but it also seems sensitive enough to detect differences from the environment, notably, between an OM and a simulated session. This finding suggests that this method could be used with the aim of monitoring appropriately the loads in different situations in high-level basketball players.

In conclusion, the data of this study show that OM promoted greater internal loads as compared with those promoted by an SM. The greater physical demands associated with competition environments might explain these findings. Additional data are required to investigate the impact of physical demands related to OM and SM on neuro-immune-endocrine responses. Monitoring internal loads using session-RPE and hormonal responses might be useful to optimize the training process.

Practical Applications

The results of this study suggest that a greater magnitude of stress was experienced by athletes during OMs. The better understanding of stress responses to sports training and competitions might be useful for planning future loads and proper recovery. The precise monitoring of these responses during competition might help the strength and conditioning coaches to plan appropriate loads maximizing recovery and performance. Coaches should be aware that it is difficult to replicate the precise training loads experienced during competitive matches. Therefore, when the aim is to plan training sessions that mimic competition demands, it is imperative to create strategies to improve the similarity between training sessions and competition. Strategies that could be used during simulated training sessions that induce changes in affective states could include creating unfavorable situations relating to advantages in score to the opponents, short-term competitions within team athletes with adapted rules (i.e., number of players, reduced time), and participation of official referees during some match simulations. These types of strategies could expose the athletes to greater psychological and physiological demands by using a competition style approach. Additionally, monitoring internal loads using session-RPE and hormonal responses might be useful to identify whether demands from these simulations are in accordance with the official competitive outputs.

Acknowledgments

We would like to thank the Fundação de Amparo à Pesquisa do Estado de São Paulo (São Paulo Research Foundation, process, 2008/10404-3) for funding this research. We also wish to acknowledge the committed participation of all the basketball players and research support staff involved in this study, particularly that of Fernando Marques (in memoriam).

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